

JONATHAN NADEAU

Data Analyst

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Portfolio Website: nadeaujonny.github.io

SUMMARY

Mathematics graduate (UCSB) and Google Data Analytics certified analyst, sole developer of three internal data tools shipped at a defense and telecom electronics manufacturer. Experience spans data extraction and validation from undocumented legacy systems, Python ETL and application development, live vendor API integration, cost-optimization analysis, and SQL and BI reporting, backed by 1,500+ hours communicating quantitative findings to non-technical audiences.

TECHNICAL SKILLS

SQL & Databases: Oracle SQL, BigQuery, MySQL, SQLite · CTEs, window functions, stored procedures, DDL/DML

Python & R: pandas, NumPy, scikit-learn, Streamlit, pytest, openpyxl, matplotlib · R (tidyverse)

Spreadsheets & BI: Excel (Power Query, Power Pivot, DAX, dynamic arrays, Solver), Power BI, Tableau, Looker Studio

Data Engineering: ETL pipelines, REST API & OAuth2 integration, incremental refresh, data validation, audit logging, Git/GitHub, GitHub Actions

PROFESSIONAL EXPERIENCE

Data Analyst Intern | Escape Communications

July 2026 – Present | Torrance, CA

Sole developer of three internal data tools for the engineering and procurement functions of a defense and telecom hardware manufacturer.

- **Failure Analysis Search Tool:** Reverse-engineered an undocumented legacy Bugzilla instance and extracted a complete 4,038-record hardware failure corpus (2007–2026) with full comment history via REST API; audited data quality across all records, proved the structured failure fields unusable as signal (80%+ blank or defaulted), and redirected the design toward the free-text comment threads where the engineering knowledge lived.
- Built a dependency-free Python pipeline indexing 2,245 PCB reference designators out of free text, hand-validated per prefix, eliminating a 68–71% false-positive rate on the worst values; shipped the Streamlit search app with component-level lookup, TF-IDF similarity, and watermark-based incremental refresh, with full handoff documentation (runbook, user guide, append-only build log) enabling zero-developer operation.
- **BOM Checker (shipped company-wide):** Designed, built, and shipped a standalone Windows BOM-auditing tool (Python, pandas, openpyxl, PyInstaller; 427 automated tests) that flags 16 classes of structural and parametric defects in raw PADS Logic BOM exports without modifying a single cell, replacing hours of manual line-by-line review with a filterable defect worklist generated in seconds.
- Built an offline parametric decoder for 10+ manufacturers' part-numbering schemes, cross-checking BOM-stated specs against what each part number encodes; validated against 8 production BOMs across 5 product lines (2,332 rows), surfacing 21 datasheet-confirmed spec errors, including a 100V requirement on a 16V-rated part, 17 current-rating errors, and 13 cross-board spec contradictions.
- **BOM Sourcing & Purchase Planning Tool:** Built an end-to-end Python sourcing pipeline (479 passing tests) that validates 300+ line BOM exports, pulls availability and pricing from distributors via file parsers and live REST APIs (Mouser production API, DigiKey OAuth2), and outputs a cost-minimized, fully auditable multi-vendor purchase plan.
- Live volume pricing invisible to the prior manual process cut a real 100-board parts plan from \$129.8K to \$103.8K (roughly \$26K, or 20%); availability checks flagged 55 at-risk parts ahead of the build; briefed the VP of Engineering with a one-page readout leading with the financial case.

Mathematics Instructor & Tutor | Mr. J's Academy · Wyzant · Blue Train College Prep

Sept 2022 – May 2025 | Torrance & Goleta, CA

- Delivered 1,500+ hours of instruction spanning Algebra 1 through Calculus and SAT prep across 17+ recurring clients, maintaining a 4.8-star Wyzant rating and 100% client retention; promoted to high school lead tutor at Mr. J's Academy with a 50% compensation increase.
- Diagnosed learning gaps from assessment results and designed targeted problem sets against each student's curriculum, translating abstract quantitative material for audiences at varying levels of technical depth.

Lab Technician | Escape Communications

Summers 2017 – 2023, 2025 | Torrance, CA

- Performed QA testing and calibration on power supply units, recording multimeter measurements into Excel trackers built to automatically flag out-of-tolerance values; tracked component inventory with low-count alerting and tested completed boards for functional compliance.

SELECTED PORTFOLIO PROJECTS (15 total at nadeaujonny.github.io)

- **Supply Chain Analytics** (Oracle SQL, Excel, Power BI): 180K orders; stored procedures with scheduled automation, Power Query ETL, ABC/XYZ classification, ETS forecasting, MRP simulation, and a 6-page star-schema Power BI dashboard; diagnosed a structural 42.7% on-time delivery rate.
- **Skechers FP&A Model** (Excel): Three-statement model from FY2020–FY2024 10-K filings with Bear/Base/Bull scenarios, sensitivity tables, variance analysis, and a KPI dashboard; all balance and cash-flow checks passing.
- **Customer Churn Prediction** (Python, scikit-learn, Streamlit): Compared five classifiers on 7K records, tuned via RandomizedSearchCV, explained predictions with SHAP, and deployed a scoring app reaching 0.83 AUC and 79% recall.

EDUCATION & CERTIFICATIONS

B.A. in Mathematics | University of California, Santa Barbara

Sept 2018 – March 2023

GPA 3.40 (Mathematics GPA 3.49) · 4x Dean's Honors · Minor: Secondary Mathematics & Science Education

Data Science Certificate | UCLA Extension

June 2026 – Dec 2026 (in progress)

Intro to Data Science · Exploratory Data Analysis & Visualization · Big Data Management · Machine Learning Using Python

Google Data Analytics Professional Certificate | Coursera

Oct 2025 – Jan 2026 · 100% average, all nine courses

Single Subject Mathematics Teaching Credential Coursework | CSU Long Beach

Aug 2024 – Oct 2025 · GPA 4.00